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Roll No

EC-7013 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Neural Networks
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Derive the properties of the mutual information between two continuous valued random values X and Y.
2. Develop a perceptron for the AND function with binary inputs and bipolar targets without bias up to 2 epochs.
3. a) What are the Network pruning techniques? Explain one in details.
b) Briefly discuss unconstrained optimization techniques.
4. Realize the McCulloch-Pitts neuron model for NAND gate and NOR gate.
5. Describe back propagation? Explain the back propagation algorithm with the help of one-hidden layer feed forward network.
6. What is a Neural network? Explain network architecture with help of diagram?

OR

The four possible input patterns are defined by (0, 0), (0, 1) (1, 1) and (1, 0) which represent the cyclically ordered corners of a square. Construct the interpolation matrix for the resulting radial basis function network.

7. Describe briefly:
 - a) Kalman filter
 - b) Kullback-leibler divergence
 - c) Real time recurrent learning
8. Write short notes on (any two):
 - i) State space model
 - ii) Regularization theory
 - iii) Boltzmann learning
 - iv) Reference adaptive control
